

MOODCLOSET

A Multi-Stage AI System for Personalized Outfit Generation

Clothing Classification, Skin Undertone Detection & Mood-Based Recommendations



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Project Description

MoodCloset is a hybrid artificial intelligence system designed to generate personalized outfit recommendations. The system analyzes a clothing image, extracts dominant colors, detects the user's skin undertone, and incorporates the user's selected mood to recommend visually harmonious and emotionally appropriate outfits.

Key Features:

- Hybrid multi-stage AI pipeline
- Lightweight MobileNetV2 model
- Skin undertone-aware recommendations
- Mood-based color psychology
- Explainable rule-based decision logic



Interface

Methodology

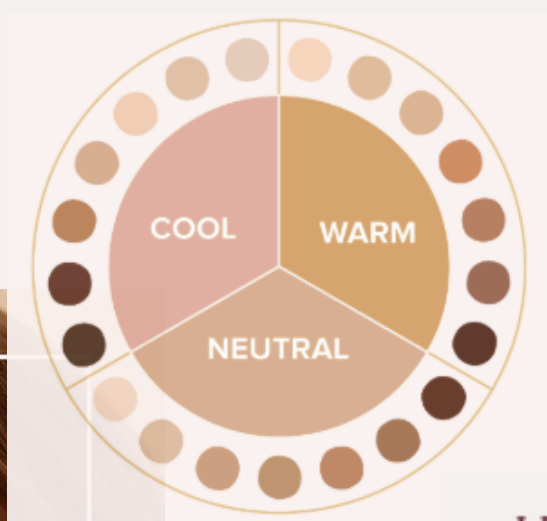
Step1: Clothing Classification
MobileNetV2 with transfer learning was used to classify clothing images into nine categories.

Step2: Dominant Color Extraction
K-Means clustering ($k = 3$) was applied to extract the dominant colors from the clothing item.

Step 3: Skin Undertone Detection
Skin undertone was detected using rule-based RGB analysis and classified as Warm, Cool, or Neutral.

Step 4: Mood-Based Mapping
User-selected moods were mapped to predefined color palettes based on psychological color theory.

Step5: Final Outfit Recommendation
The final recommendation combines clothing type, extracted colors, skin undertone, and mood.



- Happy
- Calm
- Confident
- Relaxed

Results

- Training Accuracy: 89.55%
- Validation Accuracy: 82.40%
- High precision across most clothing categories
- Positive human evaluation of outfit relevance

Conclusion

MoodCloset successfully integrates deep learning, color science, and psychological mood mapping into a unified system for personalized outfit generation. The results demonstrate that combining emotional and physical user attributes significantly improves recommendation quality compared to traditional systems.

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